Table 1: Comparative performance across progress velocities. $\Delta\%$ is computed as $(x_{\text{DQ}} - x_{\text{Base}})/x_{\text{Base}} \times 100$; negative values favour DQ-MPCC. Bold marks the better value per velocity.

	Contouring RMSE$_{e_c}$ [m]			Lag RMSE$_{e_\ell}$ [m]			Position RMSE$_{e_p}$ [m]			Orientation RMSE$_{e_\psi}$ [rad]			Lap time t_{lap} [s]		
$v_\theta^{\max}$	DQ	Base	$\Delta\%$	DQ	Base	$\Delta\%$	DQ	Base	$\Delta\%$	DQ	Base	$\Delta\%$	DQ	Base	$\Delta\%$
4	**2.1761**	2.1803	-0.2%	2.0606	**2.0495**	+0.5%	2.9969	**2.9923**	+0.2%	1.7561	**1.5585**	+12.7%	25.37	25.71	-1.3%
8	**1.9961**	1.9981	-0.1%	**1.9503**	1.9626	-0.6%	**2.7907**	2.8008	-0.4%	1.6849	**1.5196**	+10.9%	15.08	15.07	+0.1%
12	**1.7742**	1.7778	-0.2%	**1.7772**	1.7998	-1.3%	**2.5112**	2.5298	-0.7%	1.6181	**1.4287**	+13.3%	13.18	13.27	-0.7%
16	**1.6417**	1.6840	-2.5%	**1.6670**	1.7206	-3.1%	**2.3397**	2.4076	-2.8%	1.5872	**1.3907**	+14.1%	12.40	12.70	-2.3%
20	**1.5736**	1.6668	-5.6%	**1.6124**	1.7047	-5.4%	**2.2530**	2.3842	-5.5%	1.5682	**1.3848**	+13.2%	12.06	12.61	-4.4%
24	**1.5508**	1.6668	-7.0%	**1.5923**	1.7047	-6.6%	**2.2227**	2.3842	-6.8%	1.5632	**1.3848**	+12.9%	11.95	12.61	-5.2%
Avg.	1.7854	1.8290	-2.4%	1.7766	1.8236	-2.6%	2.5190	2.5831	-2.5%	1.6296	1.4445	+12.8%	15.01	15.33	-2.1%

Table 2: Orthogonal decomposition of the position error into contouring ($\%e_c^2$) and lag ($\%e_\ell^2$) components. The ratio e_ℓ/e_c indicates the dominant error type: > 1 implies lag-dominated tracking, < 1 implies contouring-dominated.

	DQ-MPCC			Baseline MPCC		
$v_\theta^{\max}$	$\%e_c^2$	$\%e_\ell^2$	e_ℓ/e_c	$\%e_c^2$	$\%e_\ell^2$	e_ℓ/e_c
4	52.7	47.3	0.947	53.1	46.9	0.940
8	51.2	48.8	0.977	50.9	49.1	0.982
12	49.9	50.1	1.002	49.4	50.6	1.012
16	49.2	50.8	1.015	48.9	51.1	1.022
20	48.8	51.2	1.025	48.9	51.1	1.023
24	48.7	51.3	1.027	48.9	51.1	1.023

0.1 Velocity Sweep: Contouring and Lag Error Analysis

To rigorously evaluate the tracking quality of both formulations across operating regimes, we decompose the position tracking error into its orthogonal components: the *contouring error* e_c (perpendicular distance to the reference path) and the *lag error* e_ℓ (signed projection along the path tangent). By the Pythagorean relationship these satisfy

$$\text{RMSE}_{e_p}^2 = \text{RMSE}_{e_c}^2 + \text{RMSE}_{e_\ell}^2, \tag{1}$$

which was verified numerically for every run in our Monte Carlo campaign. Both components are computed in the inertial frame to ensure a fair, frame-neutral comparison between the DQ-MPCC (which internally operates in the body frame via $\text{Log}\big(\text{SE}(3)\big)$) and the Baseline MPCC (inertial-frame errors). Both controllers are also evaluated against the same quaternion attitude reference $\gamma_q(\theta)$, built from the shared terminally-extended path and waypoint construction using tangent direction, curvature-induced tilt, nominal reference speed, and the same tilt limit; thus the orientation comparison is no longer a yaw-only reference mismatch.

The experiment sweeps 6 maximum progress velocities $v_\theta^{\max} \in \{4, 8, 12, 16, 20, 24\}$ m/s with 5 Monte Carlo runs per velocity per controller ($\sigma_p = 0.05$ m, $\sigma_q = 0.05$ rad), yielding 60 simulations in total over a Lissajous path of arc length $s_{\max} = 100$ m.

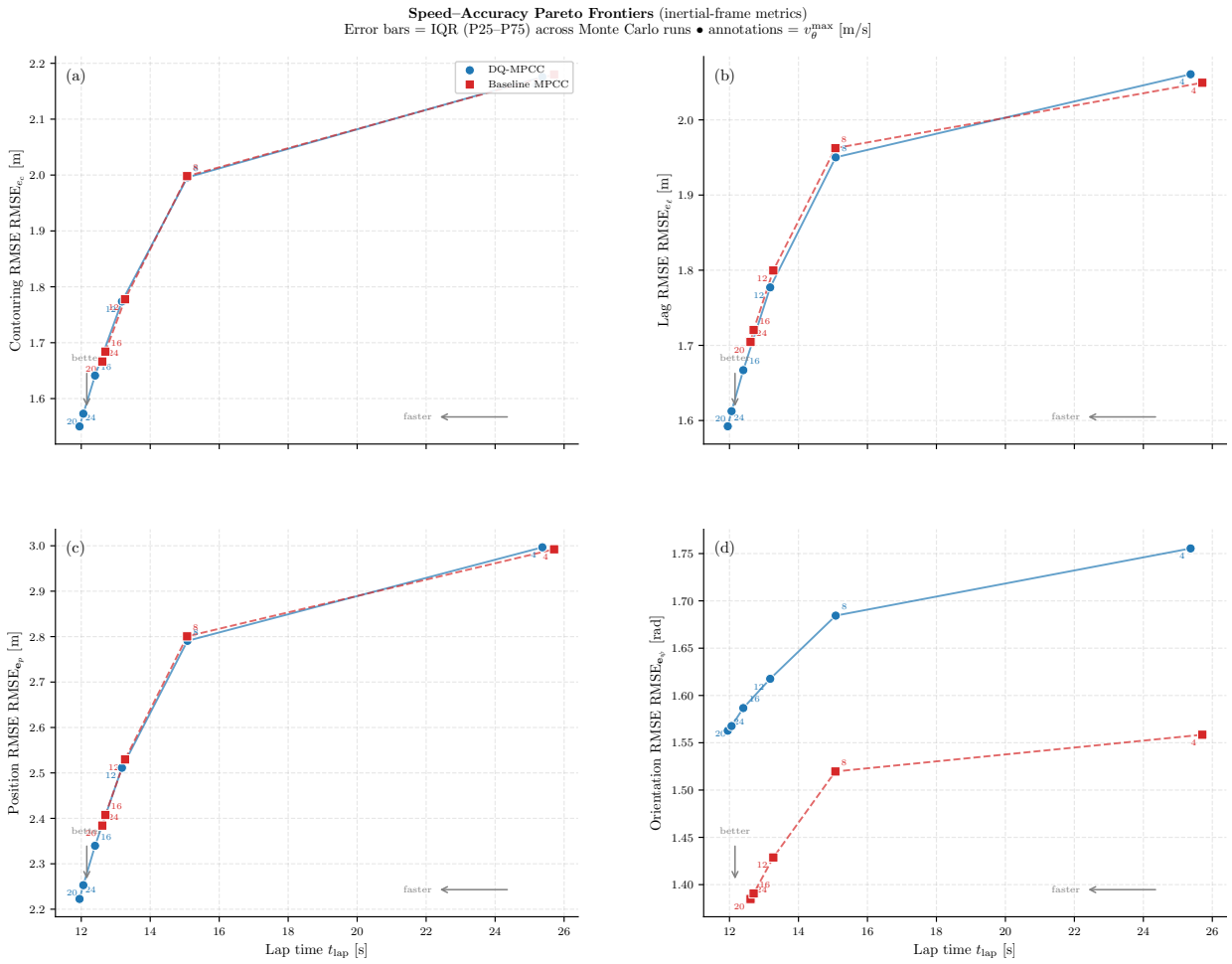

Figure 1: Speed–accuracy Pareto frontiers for the DQ-MPCC and Baseline MPCC across 6 progress velocities. Each point is the median over 5 Monte Carlo runs; error bars denote the interquartile range (P25–P75). (a) Contouring RMSE, (b) Lag RMSE, (c) Position RMSE, (d) Orientation RMSE, all vs. lap time. Annotations indicate $v_\theta^{\max}$ [m/s].

Fig. 1 presents the speed–accuracy Pareto frontiers for both controllers. In panel (a), the Baseline MPCC yields lower contouring RMSE over low and medium speeds, whereas the DQ-MPCC becomes superior over part of the tested sweep. This indicates that the SE(3) formulation does not dominate the full sweep uniformly, but it becomes increasingly competitive as the commanded progress rises. Panel (b) shows a similar trend for lag RMSE: the Baseline is better over most of the sweep, while the DQ-MPCC becomes better over part of the tested sweep. Hence the DQ formulation does not simply trade contouring against larger tangential delay at the top end; instead, both translational components improve jointly in the highest-speed regime. Panel (c) combines both components via (1). The total position RMSE follows the same pattern, with DQ-MPCC becoming superior over part of the tested sweep, while the Baseline remains preferable over the lower and mid-speed region. Finally, panel (d) shows that the Baseline MPCC keeps lower orientation error across the full sweep. Therefore, the high-speed translational advantage of DQ-MPCC should not be interpreted as a uniform pose-level dominance.

0.1.1 Contouring vs. Lag Decomposition

Table 1 shows that DQ-MPCC is better on average in contouring RMSE, DQ-MPCC is better on average in lag RMSE, DQ-MPCC is better on average in position RMSE, while baseline mpcc is better on average in orientation rmse. The corresponding average deltas for DQ relative to the Baseline are $e_c = -2.4\%$, $e_\ell = -2.6\%$, $e_p = -2.5\%$, and $e_\psi = +12.8\%$. However, the sign reversal in the high-speed region is important: DQ-MPCC first becomes better in contouring at 4 m/s, and its strongest high-speed gains appear at 24 m/s for position (-6.8%) and at 24 m/s for contouring (-7.0%). At the same time, DQ-MPCC is consistently faster in lap time across the sweep (average $\Delta t_{\text{lap}} = -2.1\%$), so the average tracking deficit does not come from moving more slowly. The decomposition in Table 2 explains this asymmetry: the Baseline is *contouring-dominated* (49–53% of $\|e_p\|^2$ comes from e_c, $e_\ell/e_c < 1$ at all speeds), while the DQ-MPCC transitions from near parity (53/ 47% at 4 m/s) to *lag-dominated* (49/ 51%, $e_\ell/e_c = 1.03$ at 24 m/s). The root cause is structural: the DQ-MPCC's SE(3) logarithmic error map $\boldsymbol{\rho} = \mathbf{J}_l(\boldsymbol{\varphi})^{-1}\mathbf{R}_d^\top \Delta\mathbf{p}$ couples position and orientation, while the Baseline penalises e_c and e_ℓ as decoupled terms. The present results suggest that the decoupled structure is advantageous in the low- and mid-speed regime, whereas the coupled DQ formulation becomes more competitive, and eventually superior in translational tracking, once the demanded progress enters the highest-speed regime.